

Alberta Buck - Wormhole (DRAFT v0.2)

Identity-Preserving Transaction Graph Privacy

Perry Kundert

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Alberta Buck's identity system provides credential unlinkability and identity privacy, but the Ethereum transaction graph remains public: balances, transfers, and counterparty relationships are visible to all. This document proposes a *wormhole* mechanism – adapted from EIP-7503¹ – that provides transaction graph privacy while preserving the identity guarantees that distinguish BUCK from anonymous cryptocurrencies.

The wormhole uses fixed-denomination burn-and-mint transfers: Alice sends BUCKs to a provably unspendable address (which holds a valid, identity-registered credential with the same M as her source account), then proves ownership via a SNARK that internally verifies a Chaum-Pedersen identity binding, and mints the same amount at a fresh address. The mint uses `_mint_noincrease` – no change to `totalSupply`, no Jubilee distortion, no PID controller interference. Fixed denominations defeat amount correlation; freely chosen BUCK-ages at the destination defeat age correlation and absorb fractional remainders, eliminating the need for non-standard "change" amounts.

The wormhole exposes a deeper primitive: *Identity Guardians* – identity-based authorization that decouples ECDSA key possession from the right to act on an account. Burn addresses (no ECDSA private key) are handled via `approveFor` with Schnorr proof of the `alt_bn128` identity key. The same mechanism enables cold wallet management, account recovery after key loss (a designated identity activates after a period of inactivity), and cosigned transfers (a spouse or partner must co-approve amounts above a threshold).

The key contribution over EIP-7503 and existing privacy mechanisms (Tornado Cash, Zcash, Aztec) is *identity continuity*: the SNARK proves that a legitimately identified person burned and minted, without revealing who. Regulators get identity recoverability at the destination address via the standard ElGamal proof chain. Users get transaction graph privacy. (PDF, Text)

¹EIP-7503: Zero-Knowledge Wormholes. Kambakhsh, K. et al., 2023. A protocol-level mechanism for private ETH transfers via proof-of-burn, using ZK-SNARKs to prove that ETH was sent to a provably unspendable address and authorizing re-minting at a new address. <https://eips.ethereum.org/EIPS/eip-7503>

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1 The Problem: Transaction Graph Leakage

Alberta Buck’s identity architecture (Identity, Examples) solves identity privacy: no observer can determine who owns an address, and no observer can link two credentials to the same person. But all transactions occur on a public Ethereum ledger. Balances, transfer amounts, counterparty addresses, and timing are visible to everyone.

Standard chain analysis can reconstruct the flow of funds between addresses. Even without knowing *who* owns each address, the *pattern* of transactions reveals economic relationships: which addresses transact together, how funds flow through the system, and which addresses likely belong to the same entity (clustering heuristics).

This is a separate problem from identity privacy, requiring separate cryptographic machinery. The identity layer (Chaum-Pedersen, PS signatures, ElGamal) ensures no one learns your name from an address. A privacy layer must ensure no one learns your address from your transactions.

2 Economic Model: Demurrage-Compatible Privacy

BUCK has economic properties that both complicate and enable a wormhole mechanism.

2.1 Demurrage and Jubilee

Every BUCK balance is subject to demurrage: 2% per year, linear (not compounding), computed lazily at transfer-out time. The contract stores two values per account: the balance, and the *BUCK-age* – a cumulative balance-time accumulator (in BUCK-years) updated at every balance-changing event. Demurrage owed at transfer-out is:

$$\text{demurrage} = \text{BUCK-age} \times 2\%$$

For a constant balance B held for T years: $\text{BUCK-age} = B \times T$, so $\text{demurrage} = 0.02BT$. After 50 years the entire balance is consumed. The *net available* at any moment is:

$$\text{net_available} = B - 2\% \times \text{BUCK-age}$$

The Jubilee account grows as a function of `totalSupply` and the demurrage rate. It represents the accumulated demurrage across all BUCK holders, redistributed to the BUCK issuers who pledged assets to create the BUCK liquidity in circulation.

2.2 Why `_mint_noincrease` Works

A standard burn-and-mint wormhole (as in EIP-7503) burns ETH at one address and mints it at another. For ETH, this is supply-neutral because the burn reduces supply and the mint restores it.

If we used standard `_mint` (which increments `totalSupply`), the total nominal supply would increase – the burn address keeps its balance AND new BUCKs are created.

The solution: `_mint_noincrease`.

Standard `_mint`:

```
_balances[to] += amount;  
_totalSupply += amount;           // supply increases
```

```
_mint_noincrease(to, amount, buckAge):
```

```

_balances[to] += amount;
_buckAge[to]   = buckAge;    // set arbitrary BUCK-age
// _totalSupply unchanged    // supply stays constant

```

The critical detail: `_mint_noincrease` sets *both* the balance and the BUCK-age at the destination. The wormhole caller is free to choose any BUCK-age for the destination account, provided the SNARK proves the conservation constraint (see Fixed-Denomination Wormholes below).

After a wormhole transfer:

- `sum(balances) = totalSupply + dead_burn_sum`
- The difference (`dead_burn_sum`) is auditable from `WormholeTransfer` event logs
- Jubilee growth is computed from `totalSupply - unaffected`
- The BUCK_K PID controller uses `totalSupply - unaffected`
- The burn address's balance is phantom: it exists in state but is economically inert

The wormhole user creates a "ghost balance" – the burned balance is permanently lost, but must continue to appear to all outside observers to be a valid, potentially usable sum of BUCKs. It sits in contract storage, accruing phantom demurrage that is never collected. Over 50 years, it would have reached zero anyway; instead, it simply never participates in the economy again.

2.3 Supply Accounting

Metric	Before wormhole	After wormhole	Change
<code>totalSupply</code>	S	S	none
<code>sum(balances)</code>	S	S + burn_amount	+burn_amount
Live circulating	S	S	none
Dead (burn addresses)	0	burn_amount	+burn_amount
Jubilee growth rate	f(S)	f(S)	none
PID controller input	S	S	none

The only observable on-chain effect: `sum(balances)` exceeds `totalSupply` by exactly the cumulative wormholed amount. This is intentional and auditable.

3 Protocol Design

3.1 Fixed-Denomination Wormholes

Fixed denominations eliminate amount correlation. All wormhole transfers use standard amounts (e.g., 100, 1,000, 10,000 BUCK), so observers cannot match a specific burn to a specific mint by amount alone.

But fixed denominations create a *change problem*: Alice's net available balance rarely decomposes exactly into standard denominations. BUCK-age solves this.

3.1.1 One Burn Address per Denomination

Each wormhole certificate is backed by exactly one burn address, and each burn address produces exactly one nullifier: `sha256(MAGIC_NULL || secret)`. This is the double-spend guard: one secret, one nullifier, one certificate.

Alice decomposes her source balance into standard denominations *before burning*, transferring each denomination to a separate burn address. Each burn address gets its own identity credential (same M , fresh keys), its own `approveFor` + `approve` handshake, and its own transfer of a fixed-denomination amount.

Since `_mint_noincrease` controls both the balance and the BUCK-age at the destination, Alice can choose *any* BUCK-age for each certificate, provided the single-certificate conservation constraint holds:

$$\text{amount} - 2\% \times \text{buckAge}_{\text{dest}} \leq \text{amount} - 2\% \times \text{buckAge}_{\text{burn}}$$

That is, the destination's net-available must not exceed the burn address's net-available. Since the burn just received a fresh transfer, its BUCK-age at the moment of burn depends on how long Alice holds the balance at the source before burning (typically near-zero for a fresh transfer). The BUCK-age at the destination is freely chosen: Alice can "age" the certificate by setting a higher BUCK-age (reducing its net value) to absorb any remainder.

Source: 2,100 BUCK (fresh, net 2,100)		
Burns to separate addresses	Burn amount	Burn address
Transfer 1	1,000	addr_burn_1
Transfer 2	1,000	addr_burn_2
Transfer 3	100	addr_burn_3
Total burned	2,100	

Each burn address backs one certificate. The destination BUCK-age is freely chosen per certificate:

Certificate	Denomination	Dest BUCK-age	Dest net	Destination
1	1,000	0 (fresh)	1,000	addr_dest_1
2	1,000	5,000 (5 yrs)	900	addr_dest_2
3	100	0 (fresh)	100	addr_dest_3
Total			2,000	

Certificate 2 voluntarily "ages" the destination by setting a BUCK-age of 5,000 (equivalent to 5 years at 1,000 BUCK), reducing its net value by 100. This makes the destination look like a naturally held account rather than a fresh wormhole mint.

Each certificate is independently verifiable. No on-chain state links the three certificates to each other or to the source account. The budget constraint is enforced per-certificate: each burn address holds exactly one fixed denomination, and each certificate draws at most the net-available at *its* burn address. There is no global budget to enforce across certificates, because the global budget was enforced at the ERC-20 layer when Alice transferred fixed amounts to separate burn addresses.

3.1.2 Privacy Benefits

1. **Amount correlation defeated:** every wormhole transfer is a standard denomination (1,000 BUCK looks like any other 1,000 BUCK).
2. **Age correlation defeated:** the destination BUCK-age bears no relation to the source BUCK-age. Eve cannot match burns to mints by observing account ages.
3. **Exact decomposition:** the continuous BUCK-age parameter eliminates the need for non-standard "change" amounts that would fingerprint the source.
4. **Multiple accounts:** certificates are deposited into separate Ethereum accounts, each with its own identity credential (same M , fresh keys), further shrouding their relation to each other and to the source.

3.1.3 Step 1: Burn

Alice holds BUCKs at `addr_source` with identity credential (`pk_s`, `E_s`, `sigma'_s`). For each denomination in her decomposition, she derives a fresh credential from her Identity Fountain – same $M = m \cdot G$, fresh keys, fresh rerandomized signature – and registers it at a separate burn address.

Burn address derivation (EIP-7503 style), repeated per denomination:

```
secret_i = random 256-bit value
addr_burn_i = sha256(MAGIC_BURN || secret_i)[12:] // last 20 bytes
```

SHA-256 produces the address, not Keccak-256.

Since Ethereum derives addresses from Keccak-256(pubkey),

no ECDSA private key maps to this address.

Each `addr_burn_i` is provably unspendable.

For each burn address, Alice:

1. Registers an identity credential via `registerCredentialFor(addr_burn_i, pk_i, E_i, sigma'_i, pi_i)`
2. Completes the bilateral identity exchange via `approveFor(addr_burn_i, ...) + approve(addr_burn_i, ...)`
3. Transfers a fixed denomination: `buck.transfer(addr_burn_i, denomination)`

To every observer, these look like Alice paying several people. Each burn address has a registered identity, participates in a Chaum-Pedersen identity exchange, and receives a standard transfer. Nothing distinguishes them from normal BUCK accounts – except that they never transact outward (indistinguishable from idle wallets).

3.1.4 Step 2: ZK Proof

For each burn address, Alice constructs a SNARK. The *private witness* contains:

- | | |
|---|--|
| 1. <code>secret</code> | -- preimage of this burn address |
| 2. <code>merkle_proof</code> | -- proof that <code>addr_burn</code> has (<code>balance</code> , <code>buckAge</code>)
in Ethereum's state trie |
| 3. <code>burn_balance</code> , <code>burn_age</code> | -- balance and BUCK-age at burn address |
| 4. <code>E_burn</code> | -- ElGamal ciphertext at burn address |
| 5. <code>E_dest</code> | -- ElGamal ciphertext at destination address |
| 6. (<code>k1</code> , <code>k2</code> , <code>s1</code> , <code>s2</code> , <code>e</code>) | -- Chaum-Pedersen proof that <code>E_burn</code> and
<code>E_dest</code> encrypt the same <code>M</code> |
| 7. <code>sigma'_burn</code> , <code>sigma'_dest</code> | -- PS signatures (issuer-verified) |

The SNARK circuit internally verifies:

- `sha256(MAGIC_BURN || secret)[12:] = addr_burn` (burn address valid)
- Merkle proof against a recent state root (burn address holds `burn_balance` with `burn_age`)

- **Denomination:** `burn_balance` is a valid denomination (the burn address received exactly one fixed-denomination transfer)
- **Conservation:** $\text{amount} - 2\% * \text{dest_buckAge} < \text{burn_balance} - 2\% * \text{burn_age}$ (destination net does not exceed burn net)
- Chaum-Pedersen equations (same identity on both sides)
- PS pairing check (both credentials are issuer-signed)
- Nullifier = `sha256(MAGIC_NULL || secret)` (computed correctly)

Public inputs (what the EVM sees):

```

nullifier    = sha256(MAGIC_NULL || secret)
amount       = one of {1, 10, 100, 1000, 10000} // fixed denomination
buck_age     = chosen BUCK-age for destination // continuous parameter
destination  = addr_dest
state_root   = recent block's state root
snark_proof  = pi

```

Each burn address produces exactly one nullifier from its unique `secret`. The EVM rejects any nullifier that has already been used. Since each burn holds exactly one fixed denomination, and each certificate can draw at most the net-available at *its* burn address, the global budget constraint is enforced by the ERC-20 layer: Alice cannot burn more than she has, and she cannot mint more than she burned.

3.1.5 Step 3: Mint

```

function wormholeTransfer(
    uint256      amount,           // must be a valid denomination
    uint256      buckAge,         // chosen BUCK-age (in BUCK-years, scaled)
    address      destination,
    bytes32      nullifier,
    bytes32      stateRoot,
    bytes calldata proof
) external {
    require(validDenomination(amount));
    require(!nullifierUsed[nullifier]);
    require(isRecentStateRoot(stateRoot));
    require(identityRegistry.isVerified(destination));
    require(verifySNARK(proof, nullifier, amount, buckAge,
                        destination, stateRoot));

    nullifierUsed[nullifier] = true;
    _mint_noincrease(destination, amount, buckAge);

    emit WormholeTransfer(nullifier, destination, amount, buckAge);
}

```

The `buckAge` parameter is the freely chosen BUCK-age for the destination account. The SNARK proves that the net-available (`amount - 2% * buckAge`) does not exceed the source's net-available, and that the amount is a valid denomination. The BUCK-age is *not* required to be a denomination – it is a continuous value that absorbs the remainder when decomposing an arbitrary source balance into fixed-denomination certificates.

The destination address must have a valid identity registration. The SNARK proves (internally) that this identity shares the same M as the source. But the verifier (the EVM) sees only: "a valid proof exists, the nullifier is fresh, and the destination has a registered identity."

4 Global Commitment Pool: One Burn, Many Spends

The fixed-denomination scheme above uses one burn address per certificate. This is privacy-clean (no on-chain link between certificates) but operationally heavy: an Albertan splitting 5,000 BUCK into five 1,000-BUCK certificates pays for five identity registrations, five `approveFor` handshakes, and five fixed-amount transfers *at burn time*.

An alternative architecture, lifted from Zcash² and Aztec³, collapses this to a single burn transaction at the cost of a global commitment tree maintained by the BUCK contract. We describe it here as a forward-looking design choice; the per-burn-address scheme remains the v1 baseline because it requires no new on-chain data structures.

4.1 From Burn-Addresses to Notes

In the v1 baseline:

- Each burn is a transfer to a distinct `addr_burn_i` holding a fixed denomination
- Each spend is a SNARK against that address's (`balance`, `age`) leaf in the EVM state trie
- The state-trie leaf is the cryptographic anchor

In the note-pool design:

- A single burn appends N *note commitments* to a contract-managed Merkle tree:

$$cm_i = H(v_i, age_{0,i}, t_{burn,i}, \rho_i, \text{predicate}_i)$$

where ρ_i is per-note randomness drawn at burn time

- The burn transaction transfers $\sum v_i$ into the contract's reserve and emits $\{cm_i\}$ into the global tree
- Each spend is a SNARK proving inclusion of *some* cm_i in the tree, knowledge of its opening, and emission of a nullifier $nf_i = \text{PRF}(\text{secret} || \rho_i)$ into a global nullifier set
- The nullifier prevents double-spend; the commitment provides amount confidentiality

The structural shift is from "one anchor per certificate" (state trie leaf at `addr_burn`) to "one anchor per note" (Merkle leaf in the contract's tree). A single burn produces N anchors at once, so one transaction replaces N .

²Zcash Protocol Specification (Sapling). Hopwood, D. et al., 2018. Shielded transactions using Groth16 proofs over the BLS12-381 curve with a note commitment / nullifier / Merkle tree architecture. <https://zips.z.cash/protocol/protocol.pdf>

³Aztec Protocol. Williamson, Z. et al., 2019. Ethereum L2 rollup providing general-purpose private transactions using PLONK proofs. <https://aztec.network/>

4.2 Demurrage in the Note Model

Each note carries the demurrage state at its moment of birth:

- $age_{0,i}$: BUCK-age the burner chose to bake into note i (typically 0 for "fresh", or a positive value to absorb a fractional remainder, exactly as in the v1 free-choice mechanism)
- $t_{burn,i}$: block timestamp of the burn

At spend time t_{spend} , the SNARK constrains the destination to absorb the accumulated demurrage:

$$v_{dest} - 2\% \times age_{dest} \leq v_i - 2\% \times (age_{0,i} + v_i \cdot (t_{spend} - t_{burn,i})).$$

The note ages *while held in the pool* at the same rate that on-chain balances do. A note minted today and spent in 25 years has half its value absorbed by demurrage, exactly as a wallet balance would. The arithmetic is identical to v1; only the source of (v, age) shifts from "state-trie leaf" to "private witness bound by a Merkle inclusion proof."

4.3 Anonymity Set

v1 (per-burn-address): anonymity set is "all unspent fixed-denomination burns of this size that have not yet been redeemed." Each redemption shrinks the set by one.

Note pool: anonymity set is "all unspent notes of any size in the global tree." A small burn redeems against the same anonymity set as a large burn. Anonymity strictly improves with adoption; v1 anonymity is bounded by the fixed-denomination pool size.

4.4 Programmable Spend Predicates

Because the SNARK already evaluates a predicate against private witness data, the predicate itself can be a note-bound parameter:

Predicate	Note becomes
<code>know(secret)</code>	Bearer note (anyone spends)
<code>know(secret) AND own(pk*)</code>	Targeted note (only <code>pk*</code>)
<code>know(secret) AND own(pk*) AND t > t_unlock</code>	Time-locked targeted note
<code>know(secret) AND multisig(pk_1, ..., pk_k)</code>	Notarized escrow note
<code>know(secret) AND oracle(condition)</code>	Conditional payout note
<code>know(secret) OR (own(pk_issuer) AND t > expiry)</code>	Recoverable expiring note

The contract sees only "a valid SNARK over commitment in the tree, here is the nullifier, mint to recipient." It need not distinguish note types.

This is the substrate for *BUCK Cash* – printable bearer notes carrying QR-encoded openings of pool commitments, redeemable by any holder (or any *particular* holder, depending on the predicate). See The Alberta Buck – Cash for the full design.

4.5 Trade-offs vs. v1 Baseline

Property	v1 (per-burn-address)	Note pool
Burn-time gas	$O(N)$ registrations + xfers	$O(1)$ burn + N tree appends
Spend-time SNARK size	Merkle proof against state	Merkle proof against note tree
Anonymity set	Per-denomination cohort	Global pool
New on-chain data structure	None (uses state trie)	Note tree + nullifier set
Programmable predicates	Limited	Native
Compatibility with Identity	Direct (existing flow)	Predicate composes with CP
Compatibility with Identity Guardians	Direct	Predicate composes with Schnorr
Bearer-instrument support	No	Yes (Mode A predicate)

The dominant reason to adopt the note pool is *not* gas savings on typical wormhole transfers – it is enabling the predicate-driven note family that includes BUCK Cash, time-locked gifts, escrow instruments, and conditional payouts. The wormhole becomes one spend predicate among many, rather than the only privacy primitive.

4.6 Migration Path

The two architectures coexist cleanly:

1. v1 wormhole continues to operate against the EVM state trie
2. The contract gains a note tree and nullifier set in a v2 upgrade
3. New predicate-based products (BUCK Cash, gifts, escrow) use the note pool from inception
4. v1 wormhole certificates can be migrated voluntarily by burning to the note pool (a one-time tree append per legacy certificate)

No action is required from existing v1 users. The v1 burn-address pool remains an addressable anonymity set even after v2 rolls out.

5 Identity-Authorized Operations

Burn addresses expose a fundamental issue: the BUCK bilateral **approve** requires both parties to sign Ethereum transactions, but burn addresses have no ECDSA private key. The solution – Schnorr-authorized identity delegation – turns out to solve a much broader class of problems.

5.1 The Problem: Two Key Systems, One Authorization Model

BUCK accounts live at the intersection of two independent key systems:

Key system	Curve	Controls	Who holds it
ECDSA	secp256k1	Ethereum transactions	Address owner
Identity (BN254)	alt_bn128	Identity credentials, CP proofs	Identity holder

Normally, the same person holds both keys. But they can diverge:

- **Burn addresses:** Identity key exists (Alice generated it), ECDSA key does not (SHA-256 derivation)
- **Cold wallets:** ECDSA key exists but is offline/inaccessible
- **Lost keys:** ECDSA key is permanently lost, identity persists via issuer re-issuance
- **Smart contract wallets:** No ECDSA key at all (contracts can't sign as EOAs)

The current BUCK contract checks only `msg.sender` – the ECDSA key. If that key is unavailable, no operation is possible, even if the caller can prove they hold the *identity* that the address represents.

5.2 approveFor: Schnorr-Authorized Identity Exchange

A Schnorr proof of knowledge of sk such that $sk \cdot G = pk$ authorizes the caller to act on behalf of any address whose registered identity public key is pk . The proof demonstrates that the caller controls the identity credential – the `alt_bn128` secret key – regardless of whether they hold the ECDSA key.

Commit: $T = k \cdot G$
Challenge: $e = H(pk, T, \text{msg.sender}, \text{principal})$
Respond: $s = k - e \cdot sk \pmod{q}$
Verify: $s \cdot G + e \cdot pk \stackrel{?}{=} T$

Cost: $\sim 12,000$ gas ($2 \text{ ecMul} + 1 \text{ ecAdd} + 1 \text{ keccak256}$).

5.2.1 Why approveFor Cannot Be Used Against Normal Addresses

`approveFor` looks up the pk *already registered* at the principal address, then requires a Schnorr proof for that specific key. The security boundary is upstream, at credential registration:

- **Normal addresses:** The owner calls `registerCredential(pk, E, \sigma', \pi)`, which binds the credential to `msg.sender`. Only the ECDSA key holder can register. Since the owner chose sk privately and published only $pk = sk \cdot G$, no one else can produce a valid Schnorr proof (discrete log hardness on `alt_bn128`, ~ 128 bits).
- **Burn addresses:** No ECDSA key exists, so `registerCredential` can never be called from them. Alice uses `registerCredentialFor(addr\_burn, pk, E, \sigma', \pi)` instead. Since she generated sk_{burn} herself, she can pass the Schnorr check.

The critical invariant: **`registerCredentialFor` must reject if a credential already exists at the target address.** This prevents front-running: Eve cannot race to register her own credential at Bob's address before Bob does, because Bob registers (via `msg.sender`-gated `registerCredential`) as part of onboarding, before he can receive any transfers.

```
function registerCredentialFor(address target, ...) external {
    require(identityRegistry.getIdentityKey(target) == ZERO,
           "credential already exists"); // ← the guard
    // ... PS signature + NIZK verification, then store
}
```

The chain of custody:

Normal address: `registerCredential (msg.sender gate)`
→ only owner knows sk
→ `approveFor` rejects outsiders

Burn address: `registerCredentialFor (no prior credential)`
→ creator knows sk
→ `approveFor` works for creator

```

function approveFor(
    address principal,          // the address being represented
    address counterparty,      // who principal is approving
    uint256 amount,
    uint256[4] calldata E_counterparty, // re-encrypted identity
    uint256[3] calldata chaumPedersen,  // same-M proof
    uint256[2] calldata schnorrAuth     // proof of knowledge of sk
) external {
    // 1. Schnorr: caller knows sk such that sk * G == pk_principal
    uint256[2] memory pk = identityRegistry.getIdentityKey(principal);
    require(verifySchnorr(schnorrAuth, pk, msg.sender, principal));

    // 2. Read E_principal from storage (not calldata)
    uint256[4] memory E_principal = identityRegistry.getCredential(principal);

    // 3. Chaum-Pedersen: same M
    require(verifyChaumPedersen(chaumPedersen, E_principal, E_counterparty,
                                pk, identityRegistry.getIdentityKey(counterparty)));

    // 4. Record the identity exchange
    _receiptFragments[principal][counterparty] = keccak256(abi.encode(E_counterparty));

    emit IdentityExchange(principal, counterparty, E_counterparty);
}

```

For the wormhole burn, Alice calls three functions:

1. `registerCredentialFor(addr_burn, pk_b, E_b, '_b, _b)`
-- registers identity at burn address (NIZK proves issuer-signed)
2. `approveFor(addr_burn, alice, 0, E_burn_for_alice, CP_proof, schnorr)`
-- `addr_burn`'s half of bilateral exchange (Alice knows `sk_burn`)
3. `approve(addr_burn, amount, E_alice_for_burn, CP_proof)`
-- Alice's half of bilateral exchange (normal approve)
4. `transfer(addr_burn, amount)`
-- both `_receiptFragments` populated → bilateral check passes

To an observer, this is a normal private-to-private transfer. The `approveFor` call is indistinguishable from someone managing a cold wallet or a contract-held account.

5.3 The General Pattern: Identity Guardians

`approveFor` solves the burn address problem, but it is a special case of a more powerful primitive: *identity-based authorization*. Instead of asking "does `msg.sender` match the target address?" the contract can ask "does the caller hold an identity that is *authorized* to act on this account?"

An account can designate **guardians** – identified persons authorized to perform specific operations under specific conditions:

```

enum GuardianType { SELF, RECOVERY, COSIGNER }

struct Guardian {
    address          guardianAddress; // must have registered identity
    GuardianType     guardianType;
    uint256          activationParam; // delay (RECOVERY) or threshold (COSIGNER)
}

// mapping: principal → guardian list
mapping(address => Guardian[]) public guardians;

```

Type	Who	When active	What they can do
SELF	Same identity (same M)	Always	Full identity operations
RECOVERY	Designated person	After N days of inactivity	Redirect funds to new address
COSIGNER	Designated person	Transfers above threshold	Must co-approve large sends

5.3.1 Use Case: Wormhole Burns (SELF)

Alice holds the identity credential for `addr_burn` – same M , different keys. She proves knowledge of `sk_burn` via Schnorr and completes the bilateral identity exchange. No guardian designation needed: the contract can verify same-identity by accepting a Chaum-Pedersen proof that the caller’s credential and the principal’s credential share the same M .

5.3.2 Use Case: Cold Wallet Management (SELF)

Alice keeps her savings in a cold wallet (`addr_cold`). She derives a hot-wallet credential from the same identity (same M , fresh keys) and registers it at `addr_hot`. From `addr_hot`, she can call `approveFor(addr_cold, ...)` to manage identity exchanges for the cold wallet without ever exposing the cold wallet’s ECDSA key.

The cold wallet’s ECDSA key stays offline. Identity operations happen from the hot wallet. Funds remain in cold storage until Alice signs a transfer with the cold key.

5.3.3 Use Case: Account Recovery (RECOVERY)

Alice designates her spouse as a recovery guardian with a 90-day activation delay:

```

buck.designateGuardian(
    spouse_address,          // must have registered identity
    GuardianType.RECOVERY,
    90 days                  // activation delay
)

```

If Alice loses her ECDSA key:

1. She obtains a fresh PS signature from the issuer (re-issuance, same m , new credential)
2. She registers the new credential at `addr_new`
3. After 90 days of inactivity at `addr_old`, the recovery guardian activates
4. Spouse calls `recoverAccount(addr_old, addr_new, schnorr_proof)`

```

function recoverAccount(
    address oldAccount,
    address newAccount,
    uint256[2] calldata schnorrAuth
) external {
    Guardian memory g = getGuardian(oldAccount, msg.sender);
    require(g.guardianType == GuardianType.RECOVERY);
    require(block.timestamp > lastActive[oldAccount] + g.activationParam);

    // Verify caller is the designated guardian
    uint256[2] memory guardianPk = identityRegistry.getIdentityKey(msg.sender);
    require(verifySchnorr(schnorrAuth, guardianPk, msg.sender, oldAccount));

    // Transfer entire balance to new account
    require(identityRegistry.isVerified(newAccount));
    uint256 balance = balanceOf(oldAccount);
    _transfer(oldAccount, newAccount, balance);

    emit AccountRecovered(oldAccount, newAccount, msg.sender, balance);
}

```

Security properties:

- The 90-day delay prevents premature activation (Alice can cancel by transacting from `addr_old`)
- The guardian is identified – if the spouse acts maliciously, their identity is recoverable via the standard proof chain
- The new account must have a registered identity, maintaining BUCK’s identity invariant
- The guardian cannot redirect to their own unregistered address (`isVerified` check)

5.3.4 Use Case: Cosigned Transfers (COSIGNER)

Alice designates her spouse as a cosigner for transfers above 10,000 BUCK:

```

buck.designateGuardian(
    spouse_address,
    GuardianType.COSIGNER,
    10_000 * 10**18 // threshold in wei
)

```

For a 50,000 BUCK transfer to Bob:

1. Alice: `approve(bob, 50000, ...)` // normal identity exchange
2. Spouse: `cosign(alice, bob, 50000, proof)` // cosigner approval
3. Alice: `transfer(bob, 50000)` // proceeds with both approvals

```

function cosign(
    address principal,

```

```

    address recipient,
    uint256 amount,
    uint256[2] calldata schnorrAuth
) external {
    Guardian memory g = getGuardian(principal, msg.sender);
    require(g.guardianType == GuardianType.COSIGNER);
    require(amount >= g.activationParam); // above threshold

    uint256[2] memory pk = identityRegistry.getIdentityKey(msg.sender);
    require(verifySchnorr(schnorrAuth, pk, msg.sender, principal));

    cosignatures[principal][recipient] = amount;
    emit Cosigned(principal, recipient, amount, msg.sender);
}

```

The **transfer** function checks: if the principal has a cosigner guardian and the amount exceeds the threshold, require a matching cosignature. Below the threshold, transfers proceed normally.

Applications:

- Spousal co-authorization for large purchases
- Corporate expense policies (CFO must approve above a limit)
- Parental controls on youth accounts
- Multisig-like security without multisig wallet complexity

5.4 The Authorization Check (General Form)

The BUCK contract's authorization logic generalizes from a simple `msg.sender` check to an identity-aware policy:

```

function isAuthorizedFor(
    address actor,
    address principal,
    ActionType action,
    uint256 amount
) internal view returns (bool) {
    // 1. Direct: actor IS the principal (standard case)
    if (actor == principal) return true;

    // 2. Self-identity: actor proved same M via approveFor
    //    (verified at approveFor call time, recorded in state)
    if (selfIdentityAuthorized[principal][actor]) return true;

    // 3. Guardian policies
    Guardian memory g = getGuardian(principal, actor);
    if (g.guardianType == RECOVERY
        && block.timestamp > lastActive[principal] + g.activationParam)
        return true;
}

```

```
    if (g.guardianType == COSIGNER
        && action == ActionType.COSIGN
        && amount >= g.activationParam)
        return true;

    return false;
}
```

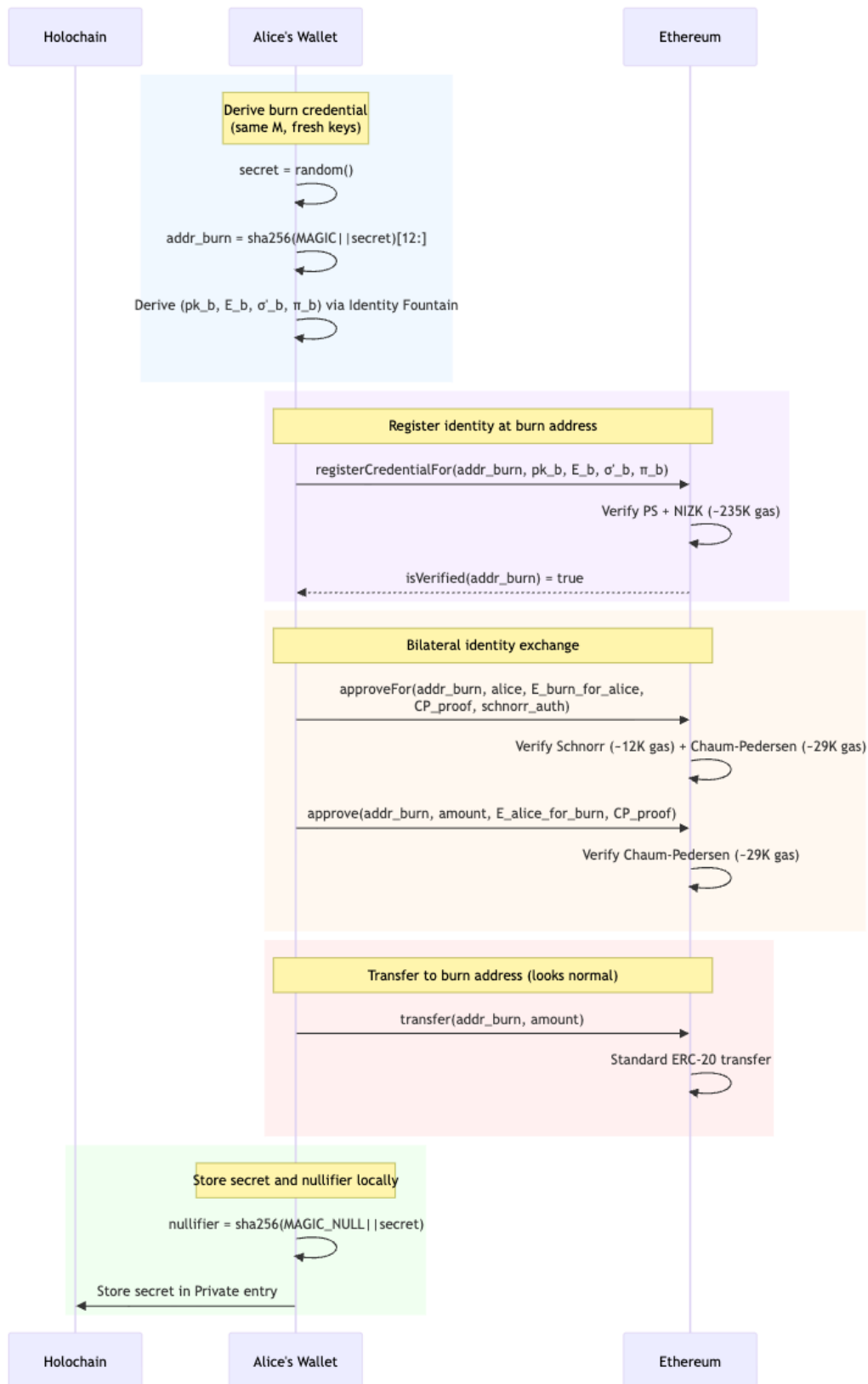
This pattern is not wormhole-specific. It extends the entire BUCK contract with identity-based authorization, using the same cryptographic primitives (Schnorr proofs, Chaum-Pedersen, ElGamal) that the identity system already provides.

DRAFT

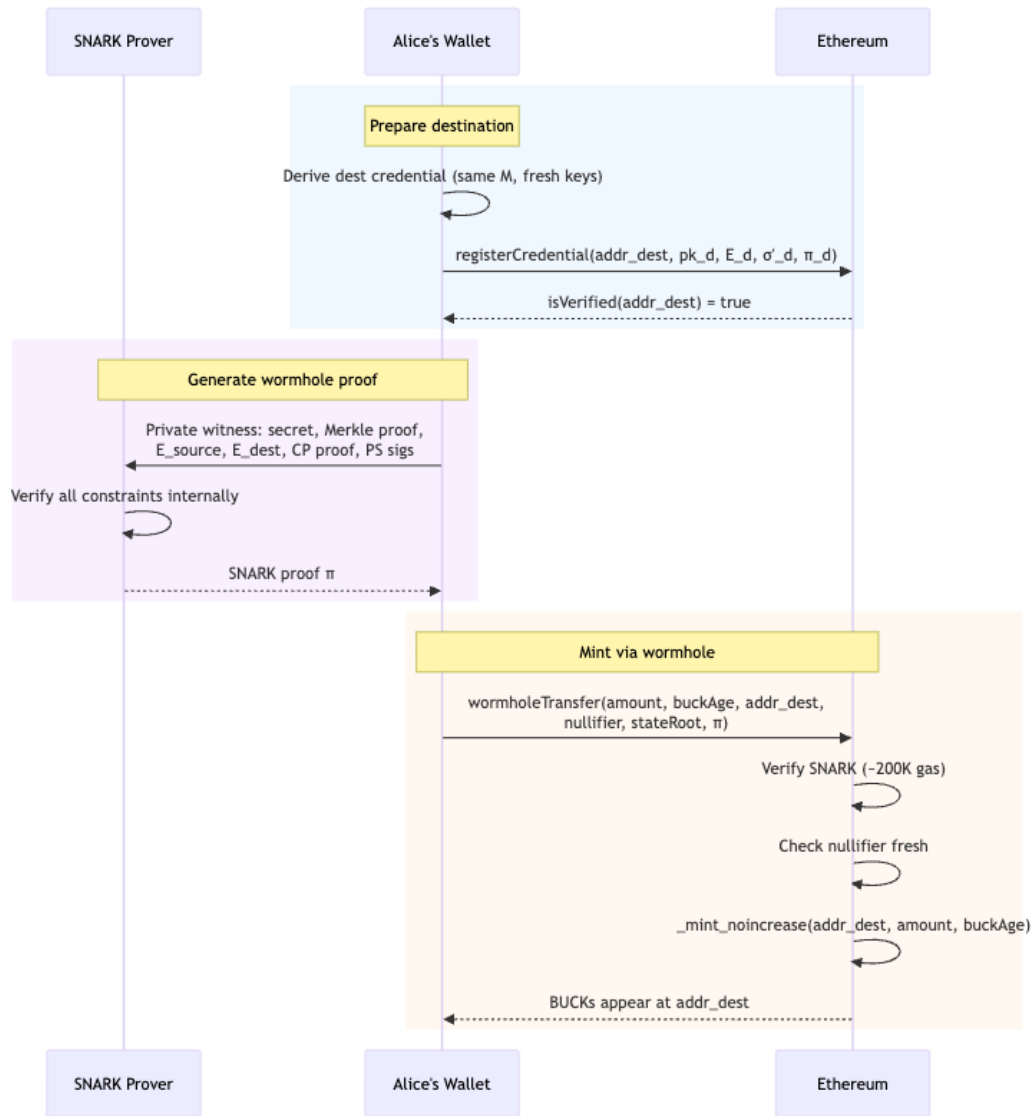
DRAFT

6 Data Flows

6.1 Wormhole Burn



6.2 Wormhole Mint



7 Worked Cryptographic Examples

These examples implement the wormhole-specific cryptographic components on the `alt_bn128` (BN254) curve. For the identity primitives they build on (PS signatures, ElGamal, NIZK, Chaum-Pedersen), see the Identity Examples document.

7.1 Setup

```

import hashlib, secrets
from py_ecc.bn128 import bn128_curve as bc
from web3 import Web3

ORDER = bc.curve_order
G1 = bc.G1; Z1 = bc.Z1; G2 = bc.G2
multiply = bc.multiply; add = bc.add
neg = bc.neg; eq = bc.eq
  
```

```

def pt(P, label=""):
    if P == Z1: return f"{label}0 (point at infinity)"
    return f"{label}({int(P[0]) % 10**6:>06d}..., {int(P[1]) % 10**6:>06d}...)"

def rand_scalar():
    return secrets.randbelow(ORDER - 1) + 1

print(f"BN254 curve order: {ORDER}")
print(f"G1 generator: {pt(G1)}")

BN254 curve order: 21888242871839275222246405745257275088548364400416034343698204186575808495617
G1 generator: (000001..., 000002...)

```

7.2 Burn Address and Nullifier

The burn address is derived via SHA-256 (not Keccak-256). Since Ethereum derives EOA addresses from Keccak-256 of a public key, no ECDSA private key can produce a SHA-256-derived address. The address is provably unspendable.

```

MAGIC_BURN = b"BUCK_WORMHOLE_BURN_V1"
MAGIC_NULL = b"BUCK_WORMHOLE_NULL_V1"

secret = secrets.token_bytes(32)

# Burn address: last 20 bytes of SHA-256
burn_hash = hashlib.sha256(MAGIC_BURN + secret).digest()
addr_burn = "0x" + burn_hash[-20:].hex()

# Nullifier: prevents double-mint
nullifier = hashlib.sha256(MAGIC_NULL + secret).digest()

# Demonstrate that SHA-256 and Keccak-256 produce different addresses
keccak_of_same = Web3.keccak(burn_hash[-20:]).hex()[-40:]

print(f"secret:      {secret.hex()[:40]}...")
print(f"addr_burn:    {addr_burn}")
print(f>nullifier:     0x{nullifier.hex()[:40]}...")
print(f"\nSHA-256 addr: {addr_burn}")
print(f"Keccak check:   0x{keccak_of_same}")
print(f"Different hash families => no ECDSA key maps to addr_burn")

secret:      f70912b6f8a863719984c103f35a665eadc7aaab...
addr_burn:    0xe3d6792b8b2a907d095f4a747608a3aba21f445c
nullifier:    0x531986bde5169c13ef5d74d2601c58ad0d119d62...

SHA-256 addr: 0xe3d6792b8b2a907d095f4a747608a3aba21f445c
Keccak check: 0xd2fc372b8e7d807983a91e7862ab056fa6e7b685
Different hash families => no ECDSA key maps to addr_burn

```

How to read the output: The SHA-256 address and the Keccak check must be different. Since Ethereum derives addresses exclusively from Keccak-256 of ECDSA public keys, no private key can produce the SHA-256-derived address. The burn address is a one-way trap.

7.3 BUCK-age Decomposition

Alice holds 2,100 BUCK (freshly received, BUCK-age 0). She decomposes into three separate burns of fixed denominations, each to its own burn address. Each certificate independently chooses a destination BUCK-age.

```

# Source account: 2,100 BUCK, freshly received (no demurrage owed)
source_balance = 2100
rate = 0.02

print(f"Source: {source_balance:,} BUCK (fresh, BUCK-age = 0, net = {source_balance:,})")

# Alice transfers fixed denominations to separate burn addresses.
# Each burn address receives exactly one denomination.
burns = [
    ("burn_1", 1000),
    ("burn_2", 1000),
    ("burn_3", 100),
]
total_burned = sum(amt for _, amt in burns)
print(f"\n Burns: {' + '.join(str(a) for _, a in burns)} = {total_burned:,}")
print(f" Remaining at source: {source_balance - total_burned:,}")

# Each certificate independently chooses a destination BUCK-age.
# Per-cert conservation: dest_net <= burn_net.
# Since each burn just received a fresh transfer, burn_age ~ 0,
# so burn_net = denomination. Alice is free to "age" the
# destination to make it look like a naturally held account.
certs = [
    (1000, 0.0),      # cert 1: brand new, full 1,000 net
    (1000, 5000.0),   # cert 2: appears 5 yrs old, 900 net
    (100, 0.0),       # cert 3: brand new, full 100 net
]

print(f"\n Wormhole certificates (one per burn address):")
print(f" {'Burn addr':>10s} {'Denom':>8s} {'Dest age':>10s}"
      f" {'Equiv yrs':>10s} {'Dest net':>10s}")
print(f" {'-'*10} {'-'*8} {'-'*10} {'-'*10} {'-'*10}")
total_net = 0
for i, (amt, age) in enumerate(certs):
    net = amt - rate * age
    total_net += net
    yrs = age / amt if amt else 0
    print(f" {'burn_'} + str(i+1):>10s} {amt:8,d} {age:10.1f}"
          f" {yrs:10.1f} {net:10.2f}")
print(f" {'-'*10} {'-'*8} {'-'*10} {'-'*10} {'-'*10}")
print(f" {'':10s} {'':8s} {'':10s} {'Total':>10s} {total_net:10.2f}")

print(f"\n Per-certificate conservation (dest_net <= burn_net):")
print(f" burn_1: 1,000 burned, 1,000 minted (age 0) <= 1,000 PASS")
print(f" burn_2: 1,000 burned, 900 minted (age 5yr) <= 1,000 PASS")
print(f" burn_3: 100 burned, 100 minted (age 0) <= 100 PASS")
print(f"\n Cert 2 voluntarily 'ages' the destination (net 900 < 1,000).")
print(f" This makes the account look like a 5-year-old holding --")
print(f" indistinguishable from a naturally aged account.")
print(f" No global budget needed: ERC-20 transfers enforce the total.")
print(f" One secret per burn, one nullifier per secret, no double-spend.")

```

Source: 2,100 BUCK (fresh, BUCK-age = 0, net = 2,100)

Burns: 1000 + 1000 + 100 = 2,100
Remaining at source: 0

Wormhole certificates (one per burn address):				
Burn addr	Denom	Dest age	Equiv yrs	Dest net
burn_1	1,000	0.0	0.0	1000.00
burn_2	1,000	5,000.0	5.0	900.00
burn_3	100	0.0	0.0	100.00

Total:	2000.00			

Per-certificate conservation (dest_net <= burn_net):

```

burn_1: 1,000 burned, 1,000 minted (age 0)   <= 1,000 PASS
burn_2: 1,000 burned,   900 minted (age 5yr) <= 1,000 PASS
burn_3:   100 burned,   100 minted (age 0)   <=   100 PASS

```

Cert 2 voluntarily 'ages' the destination (net 900 < 1,000).
This makes the account look like a 5-year-old holding --
indistinguishable from a naturally aged account.
No global budget needed: ERC-20 transfers enforce the total.
One secret per burn, one nullifier per secret, no double-spend.

How to read the output: Alice transfers three fixed denominations to three separate burn addresses. Each burn backs one certificate with one nullifier – no double-spend is possible. Certificate 2 voluntarily sets a 5-year BUCK-age, reducing its net to 900 and making the destination look like a naturally held account. An observer sees three ordinary-denomination mints with varied, plausible ages – no fingerprinting is possible.

7.4 Three Identity Credentials (Same M, Unlinkable)

Alice derives three credentials from her Identity Fountain – one for the source account, one for the burn address, one for the destination. All three share the same identity scalar $m = H(\text{identity_data})$, but every other value (keys, ciphertext, signature) is independent.

```

# Issuer key generation
x = rand_scalar(); y = rand_scalar()
X = multiply(G2, x); Y = multiply(G2, y)

# Alice's identity
identity_data = '{"name":"Alice","jurisdiction":"AB","id":"A1234567"}'
m = int(Web3.keccak(text=identity_data).hex(), 16) % ORDER
M = multiply(G1, m)

# PS signature on m
h = multiply(G1, rand_scalar())
sigma = (h, multiply(h, (x + m * y) % ORDER))

# Derive three unlinkable credentials
creds = {}
for label in ["source", "burn", "dest"]:
    t = rand_scalar()
    sigma_p = (multiply(sigma[0], t), multiply(sigma[1], t))
    sk = rand_scalar(); pk = multiply(G1, sk)
    r = rand_scalar(); R = multiply(G1, r)
    C = add(multiply(G1, m), multiply(pk, r))
    creds[label] = dict(sk=sk, pk=pk, r=r, R=R, C=C, sigma=sigma_p)

print("Three credentials derived from the same identity:")
for label, c in creds.items():
    print(f"\n {label}:")
    print(f"    pk   = {pt(c['pk'])}")
    print(f"    E.R  = {pt(c['R'])}")
    print(f"    E.C  = {pt(c['C'])}")
    print(f"    sig_1 = {pt(c['sigma'][0])}")

# Verify all decrypt to the same M
for label, c in creds.items():
    M_dec = add(c['C'], neg(multiply(c['R'], c['sk'])))
    assert eq(M_dec, M), f"{label} decrypts to wrong M"
print("\nAll three decrypt to the same M. Credentials are unlinkable")
print("but bound to the same real identity.")

```

Three credentials derived from the same identity:

```

source:
  pk    = (004481..., 120257...)
  E.R   = (609536..., 920607...)
  E.C   = (258598..., 095827...)
  sig_1 = (304976..., 397188...)

burn:
  pk    = (371444..., 238132...)
  E.R   = (462274..., 687196...)
  E.C   = (697662..., 331020...)
  sig_1 = (083145..., 148527...)

dest:
  pk    = (572550..., 137854...)
  E.R   = (867409..., 673409...)
  E.C   = (452641..., 185163...)
  sig_1 = (981341..., 098505...)

```

All three decrypt to the same M . Credentials are unlinkable but bound to the same real identity.

How to read the output: Every coordinate fragment (pk , $E.R$, $E.C$, sig_1) must be completely different across the three credentials. If any pair matched, the unlinkability property would be broken. Despite being independent, all three encrypt the same $M = m \cdot G$ – confirmed by the decryption check.

7.5 Chaum-Pedersen: Wormhole Identity Binding

The wormhole SNARK internally verifies a Chaum-Pedersen proof that the source and destination ElGamal ciphertexts encrypt the same identity point M . This is the same proof used in `approve` (see Identity Examples), but here it runs inside a SNARK rather than directly on-chain, hiding *which* addresses are involved.

```

src = creds["source"]
dst = creds["dest"]

# Alice proves E_source and E_dest encrypt the same M
# She knows sk_source (decrypts E_source) and r_dest (constructs E_dest)
k1 = rand_scalar(); k2 = rand_scalar()
T1 = multiply(G1, k1)
T2 = multiply(G1, k2)
T3 = add(multiply(dst['pk'], k2), neg(multiply(src['R'], k1)))

# Fiat-Shamir challenge
e_input = "".join(str(v) for v in [
    G1, src['pk'], dst['pk'], src['R'], src['C'],
    dst['R'], dst['C'], T1, T2, T3])
e = int(hashlib.sha256(e_input.encode()).hexdigest(), 16) % ORDER

s1 = (k1 - e * src['sk']) % ORDER
s2 = (k2 - e * dst['r']) % ORDER

# Verification (what the SNARK circuit checks internally)
v1 = add(multiply(G1, s1), multiply(src['pk'], e))
v2 = add(multiply(G1, s2), multiply(dst['R'], e))
C_ratio = add(dst['C'], neg(src['C']))
v3 = add(add(multiply(dst['pk'], s2),
    neg(multiply(src['R'], s1))),
    multiply(C_ratio, e))

print("Chaum-Pedersen wormhole identity binding:")
print(f"  Check 1 (owns source key):      {eq(v1, T1)}  PASS")
print(f"  Check 2 (knows dest randomness): {eq(v2, T2)}  PASS")

```

```

print(f" Check 3 (same M):                {eq(v3, T3)} PASS")
print(f"\nThis proof runs INSIDE the SNARK. The EVM verifier sees")
print(f"only the SNARK proof -- not which addresses or identities")
print(f"are involved.")

```

Chaum-Pedersen wormhole identity binding:

```

Check 1 (owns source key):      True PASS
Check 2 (knows dest randomness): True PASS
Check 3 (same M):              True PASS

```

This proof runs INSIDE the SNARK. The EVM verifier sees only the SNARK proof -- not which addresses or identities are involved.

7.6 Counter-Example: Different Identity at Destination

If Alice tries to mint BUCKs at an address with a *different* identity, the Chaum-Pedersen proof fails at Check 3 – the SNARK cannot be generated.

```

# Eve tries to wormhole BUCKs to an address with a different identity
m_fake = rand_scalar()
r_fake = rand_scalar()
R_fake = multiply(G1, r_fake)
C_fake = add(multiply(G1, m_fake), multiply(dst['pk'], r_fake))

k1f = rand_scalar(); k2f = rand_scalar()
T1f = multiply(G1, k1f)
T2f = multiply(G1, k2f)
T3f = add(multiply(dst['pk'], k2f), neg(multiply(src['R'], k1f)))

e_input_f = "".join(str(v) for v in [
    G1, src['pk'], dst['pk'], src['R'], src['C'],
    R_fake, C_fake, T1f, T2f, T3f])
ef = int(hashlib.sha256(e_input_f.encode()).hexdigest(), 16) % ORDER

s1f = (k1f - ef * src['sk']) % ORDER
s2f = (k2f - ef * r_fake) % ORDER

v1f = add(multiply(G1, s1f), multiply(src['pk'], ef))
v2f = add(multiply(G1, s2f), multiply(R_fake, ef))
C_ratio_f = add(C_fake, neg(src['C']))
v3f = add(add(multiply(dst['pk'], s2f),
    neg(multiply(src['R'], s1f))),
    multiply(C_ratio_f, ef))

print("Counter-example: different identity at destination")
print(f" Check 1 (owns source key):      {eq(v1f, T1f)} PASS")
print(f" Check 2 (knows dest randomness): {eq(v2f, T2f)} PASS")
print(f" Check 3 (same M):                {eq(v3f, T3f)} FAIL")
print(f"\nChecks 1-2 pass: Alice owns her key and knows the randomness.")
print(f"Check 3 fails: source encrypts m, destination encrypts m'.")
print(f"The SNARK proof cannot be generated -- identity laundering")
print(f"through the wormhole is impossible.")

```

Counter-example: different identity at destination

```

Check 1 (owns source key):      True PASS
Check 2 (knows dest randomness): True PASS
Check 3 (same M):              False FAIL

```

Checks 1-2 pass: Alice owns her key and knows the randomness.
Check 3 fails: source encrypts m, destination encrypts m'.
The SNARK proof cannot be generated -- identity laundering
through the wormhole is impossible.

7.7 Schnorr Authorization (approveFor)

A Schnorr proof of knowledge of sk authorizes the caller to act on behalf of any address whose registered identity key is $pk = sk \cdot G$. This is how Alice completes the bilateral identity exchange for the burn address (which has no ECDSA key), and how Identity Guardians authorize recovery and cosigning.

```
# Alice proves she knows sk_burn (the alt_bn128 identity key for addr_burn)
# without holding addr_burn's ECDSA key.

# --- Prover (Alice) ---
sk_burn = creds["burn"]['sk'] # burn address credential from earlier
pk_burn = creds["burn"]['pk']
caller = "0xAliceEthAddr" # msg.sender (Alice's real Ethereum address)
principal = addr_burn # the address being represented

# Commit
k = rand_scalar()
T = multiply(G1, k)

# Challenge (Fiat-Shamir: binds pk, T, both addresses)
e_input = "".join(str(v) for v in [pk_burn, T, caller, principal])
e = int(hashlib.sha256(e_input.encode()).hexdigest(), 16) % ORDER

# Respond
s = (k - e * sk_burn) % ORDER

print("Schnorr Proof of Knowledge of sk_burn")
print(f" pk_burn: {pt(pk_burn)}")
print(f" principal: {principal}")
print(f" caller: {caller}")
print(f" T: {pt(T)}")
print(f" e: {e}")
print(f" s: {s}")

# --- Verifier (on-chain) ---
# Reconstruct: s*G + e*pk should equal T
lhs = add(multiply(G1, s), multiply(pk_burn, e))
print(f"\n Verify: s*G + e*pk == T? {eq(lhs, T)}")
print(f" Authorization granted: caller may act for principal")

Schnorr Proof of Knowledge of sk_burn
pk_burn: (371444..., 238132...)
principal: 0xe3d6792b8b2a907d095f4a747608a3aba21f445c
caller: 0xAliceEthAddr
T: (826957..., 564580...)
e: 731635073087059171817706253747900590250498608299950081415358106616018304966
s: 1846597869761114896410141180132494168395149336514065531496871739923306960376

Verify: s*G + e*pk == T? True
Authorization granted: caller may act for principal
```

Counter-example: Eve tries to authorize herself for Alice's burn address without knowing `sk_burn`.

```
# Eve doesn't know sk_burn, tries with a random key
sk_eve = rand_scalar()
caller_eve = "0xEveEthAddr"

k_e = rand_scalar()
T_e = multiply(G1, k_e)

e_input_e = "".join(str(v) for v in [pk_burn, T_e, caller_eve, principal])
e_e = int(hashlib.sha256(e_input_e.encode()).hexdigest(), 16) % ORDER
```

```

# Eve responds with her own sk instead of sk_burn
s_e = (k_e - e_e * sk_eve) % ORDER

lhs_e = add(multiply(G1, s_e), multiply(pk_burn, e_e))
result = eq(lhs_e, T_e)

print("Counter-example: Eve uses wrong secret key")
print(f" Eve's sk: {sk_eve}")
print(f" pk_burn: {pt(pk_burn)} (unchanged)")
print(f" T: {pt(T_e)}")
print(f" Verify: {result} <-- FAIL")
print(f"\n s*G + e*pk != T because s encodes sk_eve, not sk_burn.")
print(f" Eve cannot call approveFor on Alice's burn address.")

Counter-example: Eve uses wrong secret key
Eve's sk: 3050787050758462618455201866989793893057872732197418173410596744076960384121
pk_burn: (371444..., 238132...) (unchanged)
T: (431812..., 652799...)
Verify: False <-- FAIL

s*G + e*pk != T because s encodes sk_eve, not sk_burn.
Eve cannot call approveFor on Alice's burn address.

```

7.8 Guardian Scenario: Account Recovery

After 90 days of inactivity, Bob (a designated **RECOVERY** guardian) can redirect Alice's funds. Bob proves knowledge of *his own* identity key, and the contract checks that Bob is registered as a recovery guardian for Alice.

```

# Bob is a RECOVERY guardian for Alice's account
sk_bob = rand_scalar()
pk_bob = multiply(G1, sk_bob)
alice_account = "0xAliceAccount"
caller_bob = "0xBobEthAddr"

# Bob generates a Schnorr proof of his own identity
k_b = rand_scalar()
T_b = multiply(G1, k_b)

e_input_b = "".join(str(v) for v in [pk_bob, T_b, caller_bob, alice_account])
e_b = int(hashlib.sha256(e_input_b.encode()).hexdigest(), 16) % ORDER
s_b = (k_b - e_b * sk_bob) % ORDER

lhs_b = add(multiply(G1, s_b), multiply(pk_bob, e_b))

print("Guardian Recovery: Bob recovers Alice's account")
print(f" Bob's pk: {pt(pk_bob)}")
print(f" Alice's addr: {alice_account}")
print(f" Schnorr ok: {eq(lhs_b, T_b)}")
print()
print(" On-chain checks (pseudocode):")
print(" 1. verifySchnorr(proof, pk_bob, msg.sender, alice_account) ")
print(" 2. guardians[alice_account] contains pk_bob as RECOVERY ")
print(" 3. lastActivity[alice_account] + 90 days < block.timestamp ")
print(" 4. → recoverAccount(alice_account, new_addr) executes")
print()
print(" If Alice is still active (check 3 fails), Bob cannot recover.")
print(" If Bob is not a registered guardian (check 2 fails), rejected.")
print(" The Schnorr proof prevents impersonation of Bob's identity.")

```

```

Guardian Recovery: Bob recovers Alice's account
Bob's pk: (867437..., 900673...)

```

```
Alice's addr: 0xAliceAccount
Schnorr ok:    True
```

```
On-chain checks (pseudocode):
1. verifySchnorr(proof, pk_bob, msg.sender, alice_account)
2. guardians[alice_account] contains pk_bob as RECOVERY
3. lastActivity[alice_account] + 90 days < block.timestamp
4. → recoverAccount(alice_account, new_addr) executes
```

If Alice is still active (check 3 fails), Bob cannot recover.
If Bob is not a registered guardian (check 2 fails), rejected.
The Schnorr proof prevents impersonation of Bob's identity.

8 Attacks and Defenses

Each attack considers what an adversary can attempt and why the protocol prevents it.

8.1 Attack 1: Amount and Age Correlation

Scenario: Eve observes a 1,000 BUCK burn from an account with BUCK-age 10,000 and looks for a mint with matching amount and age.

Defense: Fixed denominations eliminate amount correlation. BUCK-age freedom eliminates age correlation. The destination BUCK-age is freely chosen (not carried from the source), so a 1,000 BUCK mint appearing "brand new" could have come from any source regardless of that source's age. If 50 users each wormhole 1,000 BUCK in a given week, Eve sees 50 burns and 50 mints of identical amounts with *unrelated* ages. The anonymity set is all users who transacted the same denomination in a comparable time window.

8.2 Attack 2: Timing Correlation

Scenario: Alice burns at block N and mints at block $N + 5$. Eve correlates by proximity.

Defense: The protocol does not enforce timing constraints, but wallet software should introduce random delays (hours to days). SNARK certificates can be held indefinitely before redemption, allowing Alice to decouple burn and mint timing arbitrarily.

Residual risk: In early adoption (few wormhole users), timing correlation remains a statistical threat. The anonymity set grows with adoption.

8.3 Attack 3: Double-Mint (Nullifier Reuse)

Scenario: Alice tries to mint twice from the same burn.

Defense: Each burn produces a unique nullifier $N = \text{sha256}(\text{MAGIC_NULL} \parallel \text{secret})$. The contract records used nullifiers in a mapping. The SNARK proves the nullifier is correctly derived from the secret. Reuse is rejected on-chain.

Soundness: The nullifier is deterministic given the secret. Alice cannot produce a second valid nullifier for the same burn without finding a SHA-256 collision.

8.4 Attack 4: Supply Inflation

Scenario: An attacker exploits the wormhole to create BUCKs from nothing, inflating the supply.

Defense: Three layers prevent supply inflation:

1. **ERC-20 budget:** Alice can only burn what she has. Fixed- denomination transfers to separate burn addresses are standard ERC-20 operations – the balance check is enforced by the token contract.
2. **One nullifier per burn:** Each burn address has one `secret`, producing one `nullifier = sha256(MAGIC_NULL || secret)`. The EVM rejects any reused nullifier. Alice cannot mint twice from the same burn.
3. **SNARK Merkle proof:** The SNARK proves a Merkle path in Ethereum’s state trie, verifying that the burn address holds the claimed balance. The state root is a public input validated against recent block hashes. Forging a Merkle proof requires breaking SHA-256 or the SNARK’s soundness.

`_mint_noincrease` does not touch `totalSupply`, so even a successful wormhole does not inflate the economic supply metric. The only inflation is in `sum(balances)`, which is bounded by the total amount wormholed and auditable from event logs.

Critical risk: A bug in the SNARK circuit (e.g., failing to verify the Merkle proof) could enable minting without a corresponding burn. This is the highest-severity risk in any wormhole system, shared by EIP-7503. Mitigation: formal verification of the circuit, rate limiting, and a pause mechanism for the `wormholeTransfer` function.

8.5 Attack 5: Identity Laundering

Scenario: Criminal Alice deposits dirty BUCKs and withdraws clean BUCKs to an address with a different identity.

Defense: The SNARK internally verifies a Chaum-Pedersen proof that the deposit and withdrawal credentials share the same M . The same real person enters and exits. After withdrawal, the destination address has a valid identity registration. If regulators subpoena Alice at the destination address, the standard ElGamal proof chain recovers her identity.

This is the key differentiator from Tornado Cash and Zcash, which provide anonymity without identity continuity.

8.6 Attack 6: Demurrage Avoidance

Scenario: Alice wormholes BUCKs and mints at a fresh address with BUCK-age 0, hoping to reset her demurrage clock.

Defense: The SNARK enforces the conservation constraint: $\sum(\text{amount}_i - 2\% \times \text{buckAge}_i) \leq \text{net_available}_{\text{source}}$. Alice can set destination BUCK-age to 0, but the denomination must be correspondingly smaller (or she must accept more certificates with higher BUCK-ages) to satisfy conservation. She cannot create net value that exceeds what the source had after demurrage. Additionally, the burn address’s balance sits idle forever – Alice loses the full burn amount permanently.

8.7 Attack 7: Anonymity Set Degradation

Scenario: Few users adopt the wormhole, making each burn-mint pair trivially linkable.

Defense: This is an adoption problem, not a cryptographic one. The protocol is sound regardless of set size; privacy improves with usage. Mitigations:

- Wallet integration (make wormholing a default option for large transfers)

- Fixed denominations concentrate users into shared anonymity sets
- Multiple certificates per burn multiply the number of mints per burn, increasing the combinatorial difficulty of correlation

8.8 Attack 8: Burn Address Discovery

Scenario: Eve identifies burn addresses by observing addresses that receive transfers but never send.

Defense: Many legitimate accounts are idle (savings, inheritance, lost keys). Burn addresses are indistinguishable from these. They have valid identity registrations, they participated in normal `approve + transfer` flows, and their on-chain footprint is identical to any dormant wallet.

Residual risk: Statistical analysis over long time periods might identify addresses that are *always* idle as probable burn addresses. This reveals that *some* wormholing occurred, but not who did it or where the funds went.

8.9 Attack 9: SNARK Circuit Soundness

Scenario: A flaw in the SNARK circuit allows forged proofs.

Defense: This is the existential risk for any ZK-based system. Mitigations:

- Formal verification of the circuit (e.g., using Circom’s constraint checking or Lean proofs)
- Trusted setup ceremony (if using Groth16) or transparent setup (if using PLONK/STARK)
- Bug bounties and audit programs
- Rate limiting on `wormholeTransfer` (max N per block/epoch)
- Governance pause mechanism for emergencies
- Incremental deployment: deploy with restricted denominations initially, expand after battle-testing

8.10 Attack 10: Malicious Recovery Guardian

Scenario: Bob is designated as Alice’s `RECOVERY` guardian. Bob waits for Alice to become temporarily inactive (vacation, illness) and triggers recovery to steal her funds.

Defense: The `activationParam` sets a minimum inactivity period (e.g., 90 days). Any transaction from Alice resets the timer. Mitigations:

- Alice can set a long activation delay (180+ days)
- Alice can revoke or replace her recovery guardian at any time
- The recovery operation emits an event, giving Alice notice to intervene if she returns during the activation window
- Multi-guardian configurations: require 2-of-3 guardians to agree

Residual risk: If Alice is truly incapacitated for the full activation period and her guardian is malicious, funds are at risk. This is the same tradeoff as any recovery system (social recovery wallets, inheritance schemes).

8.11 Attack 11: Compromised Cosigner

Scenario: Alice designates her spouse as a **COSIGNER** for transfers above 10,000 BUCK. The spouse’s identity key is compromised (phishing, malware), and an attacker uses it to co-approve unauthorized large transfers.

Defense: The cosigner requirement is additive – an attacker needs *both* Alice’s ECDSA key (to initiate the transfer) *and* the cosigner’s identity key (to co-approve). Compromising one is insufficient.

Mitigations:

- Key rotation: the cosigner can re-derive a fresh credential (same M , new sk) and update their guardian registration
- Alert on co-approval events
- The cosigner’s Schnorr proof binds `msg.sender`, so the attacker must also control an Ethereum address (cannot replay from off-chain)

Residual risk: If both Alice’s ECDSA key and the cosigner’s `alt_bn128` key are compromised simultaneously, the two-factor protection is defeated. This is the fundamental limit of any multi-factor scheme.

8.12 Defense Summary

Attack	Primary defense	Residual risk
Amount + age correlation	Fixed denom + free BUCK-age	Low with adoption
Timing correlation	Random delays + held certificates	Moderate early on
Double-mint	Nullifier uniqueness (SHA-256)	None (deterministic)
Supply inflation	Merkle proof + <code>_mint_noincrease</code>	SNARK soundness
Identity laundering	Chaum-Pedersen in SNARK	None (math)
Demurrage avoidance	SNARK conservation constraint	None (enforced in ZK)
Anonymity set	Fixed denominations + adoption	Early adoption risk
Burn address discovery	Identical to dormant wallets	Statistical over time
SNARK soundness	Formal verification + audits	Residual (any ZK)
Malicious guardian	Activation delay + revocation	Extended incapacity
Compromised cosigner	Two-factor (ECDSA + <code>alt_bn128</code>)	Dual key compromise

9 Comparison with Existing Privacy Mechanisms

9.1 EIP-7503: Zero-Knowledge Wormholes

EIP-7503¹ proposed the burn-and-mint wormhole concept for ETH. BUCK wormholes adapt the core mechanism but diverge in four ways:

1. **Identity binding:** EIP-7503 provides pure anonymity – anyone can burn and mint. BUCK wormholes prove identity continuity via Chaum-Pedersen inside the SNARK. The same identified person enters and exits.
2. **Supply accounting:** EIP-7503 burns and re-mints ETH (supply neutral at the protocol level). BUCK uses `_mint_noincrease`, leaving `totalSupply` unchanged while the burn balance becomes phantom. This avoids interaction with BUCK-specific economics (demurrage, Jubilee, PID controller).
3. **Identity infrastructure:** BUCK burn addresses carry full identity registrations (PS signature, ElGamal ciphertext, NIZK proof), making them indistinguishable from normal accounts. EIP-7503 burn addresses are bare EOAs with no identity layer.

4. **Regulatory model:** EIP-7503 offers optional "privacy pools" compliance via association sets. BUCK provides cryptographic identity continuity – strictly stronger than social association.

9.2 Tornado Cash

Tornado Cash⁴ uses fixed-denomination deposit pools with nullifier-based withdrawal. BUCK wormholes follow a similar fixed-denomination pattern, with key additions:

- Identity-verified burns and mints (Chaum-Pedersen binding)
- BUCK-age freedom at the destination (defeats age correlation)
- Demurrage conservation enforced by the SNARK

Tornado Cash was sanctioned by OFAC in 2022 partly because it provided pure anonymity with no identity accountability. BUCK wormholes address this by ensuring identity continuity through the privacy layer.

9.3 Zcash (Sapling/Orchard)

Zcash² pioneered shielded transactions using ZK-SNARKs with a note commitment / nullifier / Merkle tree architecture. BUCK wormholes are simpler: fixed denominations with BUCK-age freedom instead of a general note-based UTXO model.

Key differences:

- Zcash is a standalone blockchain; BUCK operates on Ethereum (gas costs, EVM constraints, shared state trie)
- Zcash has no identity layer; BUCK binds every wormhole transfer to an issuer-signed identity
- Zcash provides full amount privacy (arbitrary values); BUCK uses fixed denominations with BUCK-age as the continuous parameter
- Zcash has no demurrage; BUCK's conservation constraint accounts for it

9.4 Aztec Protocol

Aztec³ brings Zcash-like shielded transactions to Ethereum as an L2 rollup with PLONK-based proofs. BUCK wormholes operate at the L1 application layer (inside the BUCK contract) rather than at the L2 infrastructure layer.

The tradeoffs:

- Aztec provides general-purpose privacy for any token; BUCK wormholes are BUCK-specific but deeply integrated with BUCK's economics
- Aztec uses universal PLONK setup; BUCK's SNARK could use the same or a simpler scheme
- Aztec has no demurrage or identity binding

⁴Tornado Cash. Pertsev, A. et al., 2019. Fixed-denomination privacy pool on Ethereum using zkSNARKs and a note commitment / nullifier scheme. Sanctioned by OFAC in August 2022.

9.5 Railgun

Railgun provides shielded ERC-20 transfers on Ethereum L1 using SNARKs. Like BUCK wormholes, it operates at the smart contract level.

Differences:

- Railgun is token-agnostic; BUCK wormholes are BUCK-specific with demurrage integration
- Railgun has no identity binding (pure privacy)
- BUCK’s identity layer enables regulatory compliance without compromising privacy for legitimate users

9.6 Comparison Summary

Feature	BUCK Wormhole	EIP-7503	Tornado Cash	Zcash	Aztec	Railgun
Identity binding	yes (CP+PS)	no	no	no	no	no
Demurrage-aware	yes (BUCK-age)	n/a	no	no	no	no
Supply accounting	_mint_noincrease	burn+mint	deposit/withdraw	UTXO	rollup	deposit/withdraw
Amount privacy	fixed denom	no	fixed denom	full	full	full
Age correlation	free BUCK-age	n/a	n/a	n/a	n/a	n/a
Anonymity set	per-denomination	all EOAs	per-denomination	shielded pool	rollup users	pool users
Regulatory compliance	identity	association	none	none	none	none
On-chain cost	~250K gas	~200K gas	~1M gas	native	rollup	~500K gas
Trusted setup	depends	yes	yes	yes (Powers)	no (PLONK)	yes
L1/L2	L1	L1	L1	L1 (own)	L2	L1

BUCK wormholes are the only mechanism combining transaction graph privacy with cryptographic identity continuity. Every other system provides either full anonymity (regulatory risk) or no privacy (identity exposure). The wormhole occupies the middle ground: privacy for users, accountability for regulators.

10 Implementation Considerations

10.1 Gas Costs

Operation	Estimated gas	Frequency
Burn (approve+transfer)	~120,000	Once per wormhole
approveFor (Schnorr)	~41,000	Once per wormhole
SNARK verification	~200,000	Per certificate
wormholeTransfer total	~250,000	Per certificate
Identity registration	~235,000	Once per address
Schnorr authorization	~12,000	Per guardian op

The SNARK verification cost depends on the proof system:

- Groth16: ~200K gas (cheapest, requires trusted setup)
- PLONK: ~300K gas (no trusted setup)
- STARK: ~500K+ gas (largest proofs, fully transparent)

10.2 SNARK Circuit Complexity

The wormhole SNARK circuit includes:

- SHA-256 hash verification (burn address and nullifier)
- Ethereum state trie Merkle proof (MPT, ~20 levels)
- Chaum-Pedersen verification (4 EC scalar multiplications)
- Conservation constraint (arithmetic on BUCK-age and amount)
- PS pairing check (3-4 pairings – expensive inside a SNARK)

The PS pairing check is the most expensive component. Alternatives:

- Replace the PS check inside the SNARK with a hash-based commitment (prove knowledge of m such that $H(m)$ matches a committed value, without the pairing)
- Use a SNARK-friendly signature scheme (e.g., EdDSA) for the identity binding, with PS verification done separately on-chain

10.3 Deployment

- Fixed denominations: 1, 10, 100, 1,000, 10,000 BUCK
- SNARK circuit: burn address + Merkle proof + Chaum-Pedersen + conservation constraint (no pairings inside the SNARK – PS verification happens at registration time, before the wormhole)
- `wormholeTransfer` contract function
- Identity Guardians (`approveFor`, recovery, cosigning)
- Wallet UI for wormhole operations and certificate management

11 Summary

BUCK wormholes provide transaction graph privacy while preserving the identity guarantees that define Alberta Buck. The mechanism adapts EIP-7503's burn-and-mint wormhole for BUCK's demurrage economics (`_mint_noincrease`, no supply inflation, Jubilee-safe) and extends it with Chaum-Pedersen identity binding inside a SNARK (same person in, same person out, provable without revealing who).

The wormhole also introduces *Identity Guardians* – a general framework for identity-based authorization that decouples ECDSA key possession from the right to act on an account. The same Schnorr proof mechanism that lets Alice manage burn addresses also enables cold wallet management, account recovery after key loss, and cosigned transfers requiring spousal or partner approval above a threshold.

Fixed denominations defeat amount correlation. Freely chosen BUCK-ages at the destination defeat age correlation and absorb fractional remainders – no non-standard "change" amounts are needed. Multiple certificates per burn, deposited into separate accounts at different times, further dissolve the link between source and destination.

The result is a privacy layer that occupies a unique position: stronger privacy than ERC-20 (transaction graph is hidden), stronger accountability than Tornado Cash or Zcash (identity continuity is cryptographically enforced), and compatible with BUCK's distinctive economic model (demurrage, Jubilee, PID stabilization).

DRAFT